

William Huang

Electrical Engineering Student | Hardware, Systems & Circuit Design

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Electrical Engineering student with experience in hardware design, PCB development, and system-level integration. Skilled in troubleshooting electrical systems using oscilloscopes, digital multimeters (DMMs), and laboratory test equipment. Experience with embedded systems, microcontrollers, and firmware development using C/C++. Hands-on background designing and building hardware systems, with a strong interest in developing innovative, real-world solutions.

TECHNICAL SKILLS & TOOLS

Hardware & Design: PCB Design (KiCad), Analog & Digital Circuits, Power Systems, Voltage Regulation, System Integration

Embedded Systems: Microcontrollers, Embedded C/C++, Firmware Development, Real-Time Systems

Interfaces: I2C, SPI, UART, USB

Tools & Software: MATLAB, Simulink, LTSpice, PSpice, Verilog, MPLAB X, Git (basic)

Lab Equipment: Oscilloscopes, Multimeters (DMM), Power Supplies

Other: Troubleshooting, Hardware Debugging, Data Analysis, English (Fluent), Mandarin (Fluent)

EXPERIENCE

Cal Poly Pomona - Bronco Star | BLADE, Team 1

Pomona, CA, September 2025 - May 2026

Power Systems Subteam Lead

- Led electrical design and integration for a 1U BalloonSat (100,000 ft operation), including power distribution and sensor interfaces
- Designed and validated a 7.4V Li-ion power system with voltage regulation and protection circuitry
- Troubleshot system-level hardware issues using oscilloscopes, DMMs, and power supplies, resolving integration and performance faults
- Performed system-level integration of sensors, power subsystems, and embedded hardware across cross-functional teams
- Developed and refined a high-altitude system from concept to functional prototype in a fast-paced, iterative environment

PROJECTS

Post-Quantum Secure Telemetry for CubeSats (NASA ORBIT) — Verilog Hardware Design Pomona, CA, Feb. 2026-Present

- Collaborated within a team to design and integrate Kyber cryptographic hardware into a secure telemetry system
- Designed modular arithmetic hardware (mod 3329) for post-quantum cryptography (Kyber)
- Implemented Barrett reduction and Montgomery multiplication for optimized performance
- Built self-checking testbenches with randomized inputs for verification
- Analyzed latency and critical path timing for performance optimization

Smartwatch Embedded Hardware Project (Kicad Self-Project)

Pomona, CA, December 2025

- Designed a complete ESP32-C3 embedded hardware system integrating power management, IMU sensors, and display interfaces.
- Created a multi-layer PCB layout with optimized component placement, routing, decoupling, and power distribution.
- Integrated I2C, SPI, and USB interfaces at the schematic and layout level.
- Verified component footprints, symbols, and pin mappings using datasheets and engineering documentation.
- Applied practical PCB design principles for manufacturability, assembly, and hardware validation.

Real-Time Fan Control System — Embedded Hardware & Control (PIC18F4620)

Pomona, CA, November 2025

- Designed a real-time embedded system integrating temperature sensing, PWM fan control, and RPM feedback
- Implemented a state-machine with manual/auto modes, dynamically adjusting duty cycle based on temperature thresholds
- Interfaced SPI (TFT), I²C sensors, ADC (photoresistor), and GPIO inputs for full system integration
- Developed interrupt-driven control using timers/counters and validated performance through debugging and testing
- Measured and monitored fan RPM and duty cycle behavior to evaluate system performance
- Developed low-level firmware in C for microcontroller-based real-time control using interrupts, timers, and PWM

EDUCATION

California State Polytechnic University Pomona

Pomona, CA

Bachelor of Science, Major in Electrical Engineering,

Expected May 2027

Cumulative GPA: 3.35

Spring 2026 GPA: 3.90

- **Relevant Coursework:**

Electromagnetism and Circuits, Advanced Circuit Analysis, Digital Logic Design, Microelectronic Devices and Circuits, Microcontrollers, Signals and Systems, Power Engineering, Control Systems

INVOLVEMENTS

Clubs: IEEE Student Member (CES), Bronco Star/Space (BLADE)